

Gitting Good at GitHub

An interactive introduction to working with Git and GitHub

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Agenda

- Module 1 - Introduction to Version Control and Git
- Module 2 - Basic Git Operations
- Module 3 - Branching and Merging
- Module 4 - Collaboration on GitHub/GitLab
- Module 5 - Advanced Git and GitHub/GitLab Features
- Module 6 - Simulated Project

Module 1 - Introduction to Version Control and Git

- What is Version Control
 - What are the benefits of Version Control?
- What is Git
 - Key Git Concepts
- What is GitHub/GitLab
 - Setting up Git with GitHub/GitLab

What is Version Control?

- Version control is a tool that allows you to track changes to files over time.
- This is most effective when working with text files but also works with other files.

What are the benefits of Version Control?

- **Track Changes.** You can see who changed what to files over time.
- **File History.** Ability to revert back to previous versions of a file.
- **Collaboration.** Supports collaborating in files.
- **AI Audit Trail.** Easily track exactly what an LLM like Claude changed during a session — and constrain its work to specific commits for full visibility and control.

What is Git?

- Git is a distributed version control system used to track changes.
 - There are centralized frameworks such as Team Foundation Server (Microsoft) and Subversion (SVN) (Apache)
- Changes are tracked locally on your computer and then synchronized with a remote repository (like GitHub/GitLab).
- Each change is saved as a snapshot, not just differences.
- Enables offline work and syncing changes when you're ready.

Key Git Concepts

- **Repository (repo).** A directory where Git stores the history of your project.
- **Commit.** A snapshot of your changes, along with a message describing what was done.
- **Branch.** A branch lets you work on changes separately from the main code.
- **Merge.** Combines commits from one branch into another.
- **Tracks Files.** Git tracks files and their file paths, not directories.

What is GitHub/GitLab?

- An online platform to store, manage, and track changes to code using Git.
- A **communication** platform.
- CI/CD
 - **GitHub Actions.** for running and deploying code.
 - **GitLab CI/CD.** for running and deploying code.
- **GitHub/GitLab Pages.** for blogs.
- Productivity tools like Issue tracking, project management, and a wiki.

Setting up Git with GitHub/GitLab

- Github
 - Install GitHub Desktop at <https://github.com/apps/desktop>
- GitLab
 - Install GitKraken at <https://www.gitkraken.com/git-client>

Module 2 - Basic Git Operations

- Ignoring Files
- Staging and Committing Changes
- When to Commit
- Undoing Changes
- Removing Files

Ignoring Files

- A `.gitignore` file is used to ignore files that you do not want to manage in source control. Some examples are:
 - Files containing secrets (passwords, API keys, etc.)
 - Locally installed package files like `.venv/`
 - Large data files (aka 2 GB csv files)

Ignoring Files (Activity)

1. Create a directory `new_folder`
2. Create a file `new_file.txt` in the `new_folder` directory.
3. Add `new_folder/` to the `.gitignore` file in the root project.
4. To the root project, add a file called `new_file.csv`
5. Add `*.csv` to the `.gitignore` file.

Staging and Committing Changes

- You can either add all files with `git add -A` or
`git add path/to/filename`
 - Tools like GitHub Desktop and VS Code's Source Control Panel are helpful to see all files
- Once a file is staged, you can then `commit` the changes with
 - `git commit -m"{insert message here}"`
- To commit all tracked files that have been modified, you can use the `-a` argument:
 - `git commit -am"{insert message here}"`

Committing Changes (Activity)

1. Make changes to the `exercises/existing_file.txt`
2. Create a file in `exercises` directory called `new_file.txt` and add text to it
3. `git commit -am "{insert your descriptive message here}"`
4. `git status`
5. `git add exercises/new_file.txt`
6. `git status`
7. `git commit -m "{insert your descriptive message here}"`
8. Check **Commit History**. You can use `git log` but GitHub Desktop or VS Code is easier

When to Commit?

- When do you commit?
 - Committing too often leads to noisy commits
 - Committing too infrequently makes it hard to find changes
- *The best time to commit is when you have completed a "thought"*
 - This could be when you finish a function, update some business logic, etc.

Undoing Changes

- `git reset HEAD~1`
 - This simply undoes the last commit with files intact
- `git checkout origin/main path/to/filename`
 - This will reset the file to what is in the main branch

Undoing Changes (Activity)

1. To undo your last commit, run `git reset HEAD~1`

You can explore other ways of changing history, but suggest using GitHub Desktop.

- `git commit --amend`
- `git revert`

Removing Tracked Files

- In the event a file gets added that you want to stop tracking, you can use the `git rm` command:
- If you want to untrack and delete the file, you can do:
 - `git rm path/to/file`
- If you want to just stop tracking it in git you can do:
 - `git rm --cached path/to/filename`
 - Add the file to your `.gitignore`
- If you want to untrack an entire directory:
 - `git rm -r path/to/directory`
- Last, you need to commit the removal change using `git commit`

Removing Tracked Files (Activity)

1. Remove `new_folder` from `.gitignore`
2. `git add new_folder/new_file.txt` file.
3. `git commit -m "New file commit message"`.
4. Remove the whole folder `git rm -r new_folder/`
5. Commit the removal `git commit -m "Remove new_folder"`

Removing Files from History

- In the event you discover a secret has been committed, it is best practices to cycle the secret (e.g. change the password, regenerate the API key, etc.) so that it is no longer valid. You can also [follow this guide for removing from the file with secrets history](#)
- It is complex and challenging to modify git history (this is a feature, not a bug).
- Be very careful to not include secrets in your branch and especially don't merge a branch with secrets into the main branch.

Module 3 - Branching and Merging

- What are Branches
- Working with Branches
- Merging Branches
- Rebasing Branches
- Merge Conflicts

What are Branches?

- A branch lets you work on changes separately from the main code.
 - For individual projects, working in `main` is "okay"
 - It is a good habit to always work in branches
 - For team based projects, always need to work in branches

Working with Branches

- Create a branch and then checkout that branch:
 - Create a branch `git branch {insert-descriptive-branch-name}`
- To change to a branch
 - `git checkout {insert-descriptive-branch-name}`
 - `git switch {insert-descriptive-branch-name}`
- Create and change to a branch
 - `git checkout -b {insert-descriptive-branch-name}`
 - `git switch -c {insert-descriptive-branch-name}`
- To delete a branch
 - `git checkout main`
 - `git branch -D {insert-descriptive-branch-name}`

Working with Branches (Activity)

1. Create a branch `git switch -c my-new-branch`
2. Delete the branch:
 1. `git checkout main`
 2. `git branch -D my-new-branch`

Merging Branches

- To merge a branch locally, you do `git merge {insert-branch-name}`
- It is common to need to merge `main` into current branch because:
 - In your Pull Request, you have merge conflicts you need to address
 - Code has been merged into main that you need for your work
- To merge main into your current branch
 - `git checkout main` (swap to main)
 - `git pull` (update main branch)
 - `git checkout {insert-branch-name}` (swap back to your branch)
 - `git merge main` (merge main into your branch)

Merging Branches (Activity)

1. Create a branch `git branch add-feature`
2. Create a file `branch_test_file.txt` and add it
 1. `git add branch_test_file.txt`
 2. `git commit -m "Add branch test file"`
3. `git checkout add-feature` (Swap to add-feature branch)
4. Merge main into `add-feature` branch
 1. `git merge main`

Merging Pros and Cons

- **Pros**

- Preserves full branch history, so you can trace what happened in team work
- Safe for collaboration because merge does **not** rewrite existing commits
- PR context stays connected to the merged work

- **Cons**

- Extra merge commits can make history feel noisy over time
- Commit history is less linear and can be harder to read quickly
- Long-lived branches can lead to more merge conflicts

Rebasing

- Rebasing replays a branch's commits on top of another branch's latest commit
- Unlike merging, rebasing rewrites commit history to produce a linear history
- To rebase main into your current branch:
 - `git switch main` (swap to main)
 - `git pull` (update main branch)
 - `git checkout {insert-branch-name}` (swap back to your branch)
 - `git rebase main` (rebase your branch on top of main)
- **Avoid rebasing branches that others are working on** — rewriting shared history can cause problems

Rebasing (Activity)

1. Create and switch to your feature branch from `main` :

1. `git checkout main`
2. `git switch -c add-feature`

2. Make your feature change and commit it:

1. Create `rebase_test_file.txt` , then run `git add rebase_test_file.txt`
2. `git commit -m "add feature work"`

3. Simulate main moving ahead, then rebase your branch:

1. `git checkout main` and commit a change to `exercises/existing_file.txt`
2. `git checkout add-feature`
3. `git rebase main`

4. Move `main` to include the rebased feature commits:

1. `git checkout main`
2. `git rebase add-feature`

Rebasing Pros and Cons

- **Pros**

- Creates a clean, linear commit history that is easier to scan
- Helps keep feature work up to date without adding merge commits
- Makes `git log --oneline` and history review simpler for many teams

- **Cons**

- Rewrites commit history, which can confuse collaborators on shared branches
- Can require force push (`git push --force-with-lease`) after rebasing remote branches
- Conflict resolution may need to happen commit-by-commit during the rebase

Merge Conflicts

- Occurs when changes in two branches **affect the same part of a file** and Git can't automatically decide which change to keep, requiring manual resolution.

```
<<<<<<< HEAD
This is the version from the main branch.
=====
This is the version from the test-sample-branch.
>>>>>>> test-sample-branch
```

Merge Conflict (Activity)

1. `git checkout main` (swap back to main)
2. Make a change to the first line in `exercises/existing_file.txt`
3. `git commit -am"Commit message for existing_file.txt"`
4. `git checkout add-feature` (swap back to add-feature)
5. Make a different change to the first line in `exercises/existing_file.txt`
6. `git commit -am"Commit message for existing_file.txt"`
7. Run `git merge main`
8. You can use the built in VS Code or GitHub Desktop to resolve the conflict.

Module 6 - Advanced Git and GitHub Features

- pre-commit
- GitHub Actions
- GitLab CI/CD

pre-commit

- pre-commit is a framework for managing and running automated Git hooks to catch and fix issues (like formatting, linting, or security checks) before code is committed, ensuring consistent code quality across teams.
- Some common pre-commit hooks are the following:
 - SQL - sqlfluff pre-commit
 - Python - ruff pre-commit
 - git secrets checker

GitHub/GitLab CI/CD

- GitHub Actions is a Continuous Integration/Continuous Deployment (CI/CD) automation tool built into GitHub that lets you automatically build, test, and deploy code based on events like pushes, pull requests, or schedule triggers.
- GitLab CI/CD is GitLab's built-in Continuous Integration/Continuous Deployment (CI/CD) platform that automates build, test, and deployment workflows using pipelines defined in a `.gitlab-ci.yml` file.

Module 5 - Simulated Project

This is a sample project for testing working with GitHub

<https://github.com/alex-antonison/git-r-done-enterprise-2026-04-30>